Exercises and Solutions

Theory of Distributions

Prepared by ...

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Exercise 1.1

Consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos[(\pi-\theta)z]}{2\sin(\pi z)}.$$

a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$. At $z = \pm i$:

$$\operatorname{Res}[F, \pm i] = -\frac{\cosh(\pi-\theta)}{4\sinh\pi}.$$

At $z = n \in \mathbb{Z}$:

$$\operatorname{Res}[F, n] = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

b) The contour $\Gamma^{(N)}$

Define

$$\begin{aligned}
\Gamma_1^{(N)}(t) &= \left(N+\tfrac{1}{2}\right)(1+it), \quad -1 \le t \le 1, \\
\Gamma_2^{(N)}(t) &= \left(N+\tfrac{1}{2}\right)(-t+i), \quad -1 \le t \le 1, \\
\Gamma_3^{(N)}(t) &= \left(N+\tfrac{1}{2}\right)(-1-it), \quad -1 \le t \le 1, \\
\Gamma_4^{(N)}(t) &= \left(N+\tfrac{1}{2}\right)(t-i), \quad -1 \le t \le 1.
\end{aligned}$$

Then $\Gamma^{(N)}$ is the closed square contour with vertices $\pm(N+\frac{1}{2}) \pm i(N+\frac{1}{2})$. It encloses the poles at $\pm i$ and all integers n with $|n| \le N$.

c) Bound on the contour integral

On $\Gamma^{(N)}$, $|z| \sim N$, so $|F(z)| = O(1/N^2)$. Since the contour has length $O(N)$, we obtain

$$\left| \int_{\Gamma^{(N)}} F(z)\, dz \right| \le \frac{C}{N},$$

for some constant C independent of N.

d) Residue theorem and Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma^{(N)}} F(z)\,dz = 2\pi i \left(\mathrm{Res}[F,i] + \mathrm{Res}[F,-i] + \sum_{n=-N}^{N} \mathrm{Res}[F,n] \right).$$

Hence

$$\left| \frac{\cosh(\pi-\theta)}{2\sinh\pi} - \sum_{n=-N}^{N} \frac{\cos(n\theta)}{1+n^2} \right| \le \frac{C}{N}.$$

Letting $N \to \infty$, we conclude

$$\frac{\cosh(\pi-\theta)}{2\sinh\pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1+n^2}.$$

This gives the Fourier series expansion, uniformly in θ.

Exercise 1.1

Consider the meromorphic function

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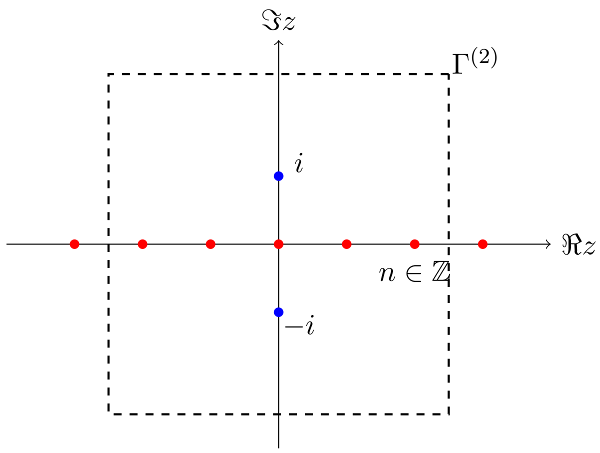
a) Poles and residues

The poles occur at $z = \pm i$ and $z = n$, $n \in \mathbb{Z}$. ...

Chart: Poles and Contour



Legend. Red dots = poles at integers, Blue dots = poles at $\pm i$, dashed square = contour $\Gamma^{(N)}$.

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d) Residue theorem and Fourier series

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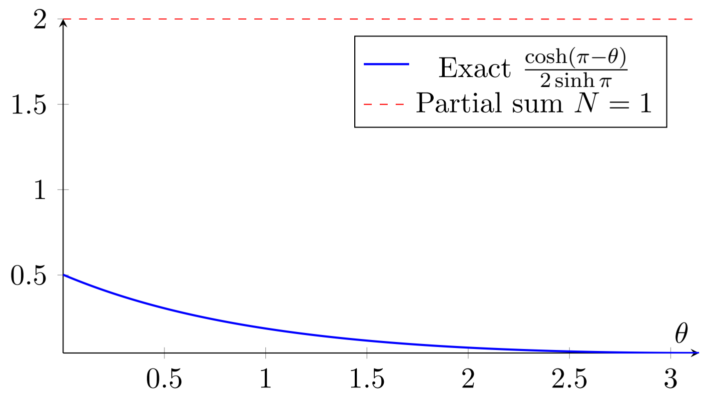
Chart: Fourier Series vs. Exact Expression

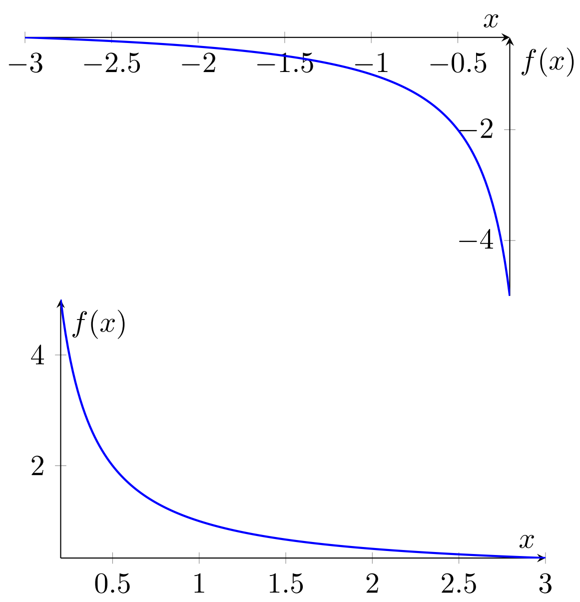


Legend. Blue curve = exact hyperbolic expression, red dashed = Fourier sum with $N = 1$, green dashed = Fourier sum with $N = 5$.

Introductory Charts: Simple Meromorphic Functions

To recall the behavior of basic meromorphic functions near poles, we show plots of $f(z) = 1/z$ and $f(z) = 1/z^2$ on the real line. Both have a pole at $z = 0$, with $1/z$ being of order 1 and $1/z^2$ being of order 2. Clearly, $(1/z)^2 = 1/z^2$.

Chart: $f(x) = 1/x$ on $\mathbb{R} \setminus \{0\}$



Legend. Function $1/x$ has a simple pole at $x = 0$. The left and right limits blow up to $\mp\infty$.

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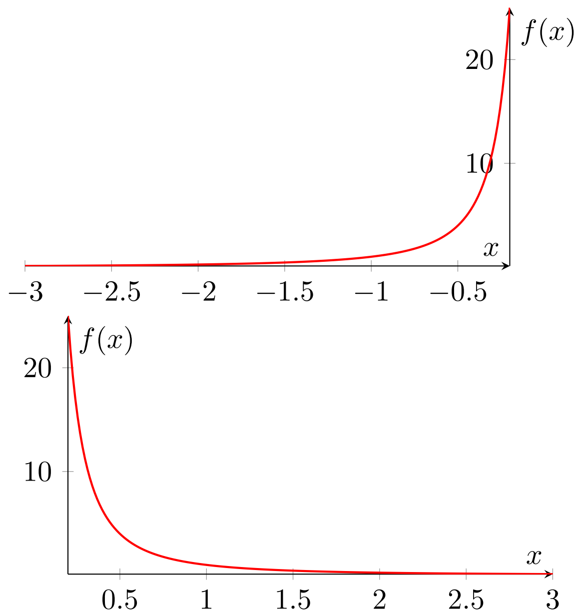
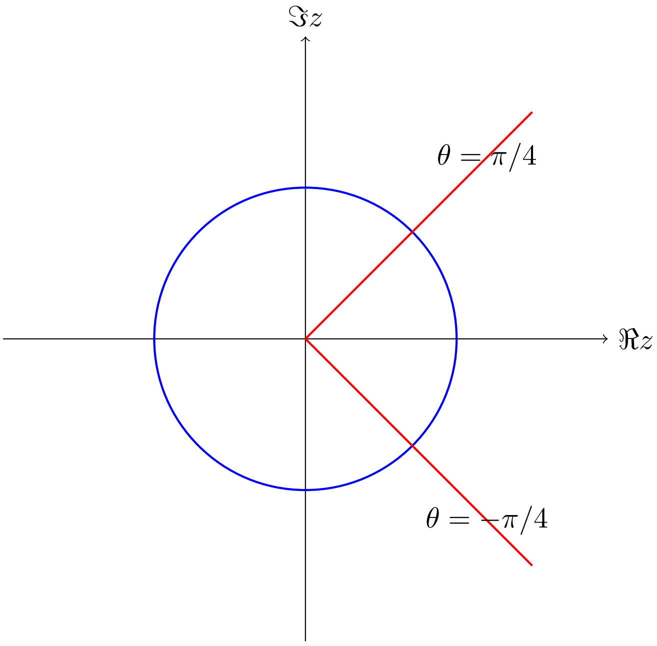
Chart: $f(x) = 1/x^2$



Legend. Function $1/x^2$ has a pole of order 2 at $x = 0$. It diverges to $+\infty$ on both sides of the pole. Clearly, $(1/x)^2 = 1/x^2$.



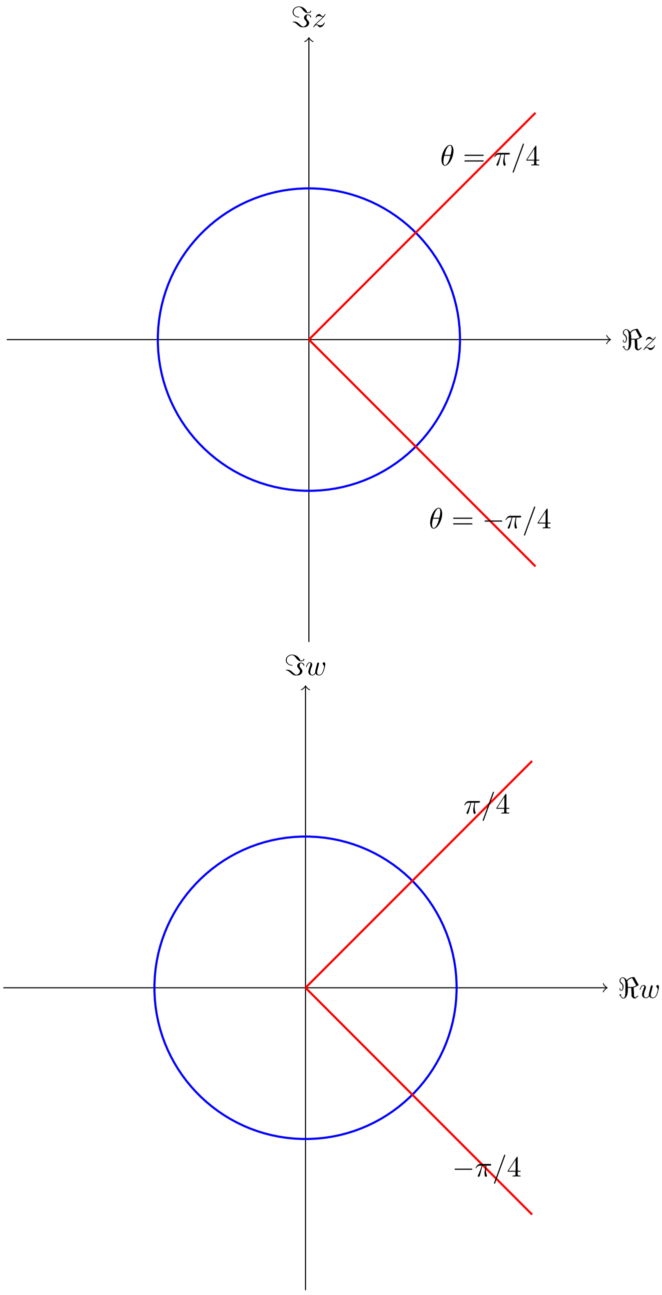
Introductory Examples: Simple Meromorphic Maps

Example 1: $f(z) = \frac{1}{z}$

Description. For $z = re^{i\theta}$ we have

$$\frac{1}{z} = \frac{1}{r}e^{-i\theta}.$$

Thus $f(z)$ inverts the magnitude ($r \mapsto 1/r$) and reflects the angle ($\theta \mapsto -\theta$). The unit circle maps to itself with reversed orientation. Lines through the origin are mapped to lines through the origin, reflected in the real axis.



Legend. Left: domain (z-plane). Right: image ($w = 1/z$). The unit circle is preserved, rays reflect across the real axis.

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Example: $f(z) = \frac{1}{z}$

Description. For $z = re^{i\theta}$ we have

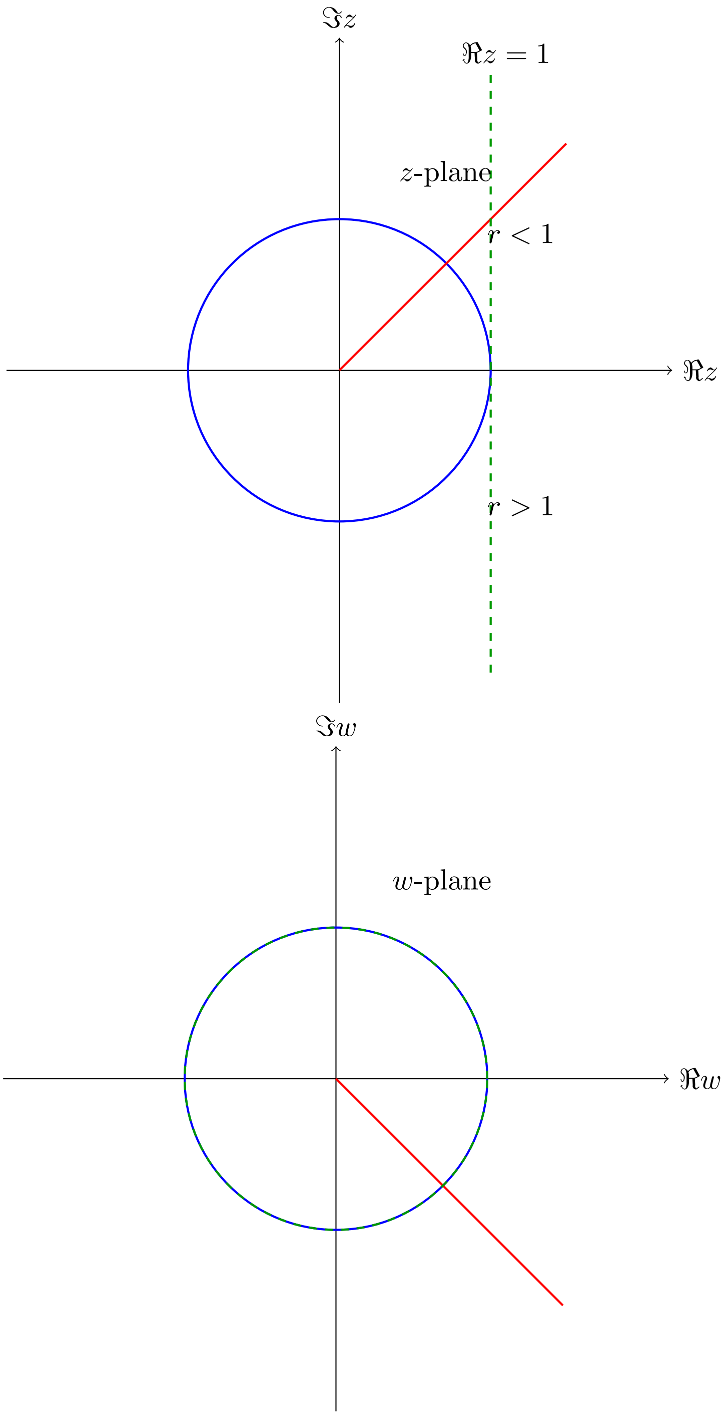
$$\frac{1}{z} = \frac{1}{r}e^{-i\theta}.$$

Thus the mapping $f(z) = 1/z$ has the following geometric effects:

- **Inversion of radii:** $r \mapsto 1/r$.
- **Reflection of angles:** $\theta \mapsto -\theta$.
- The unit circle $|z| = 1$ is invariant, but orientation is reversed.
- The interior of the unit disk $|z| < 1$ maps to the exterior $|w| > 1$, and vice versa.

- Rays through the origin map to reflected rays across the real axis.
- More generally, lines not through the origin map to circles, and circles not through the origin map to circles (inversion geometry).

Chart: Domain vs. Image under $1/z$



Legend. Blue: unit circle (invariant). Red: ray at $\theta = \pi/4$ mapping to $\theta = -\pi/4$. Green dashed: vertical line $\Re z = 1$ maps to a circle through the origin.

Example: $f(z) = \frac{1}{z+1}$

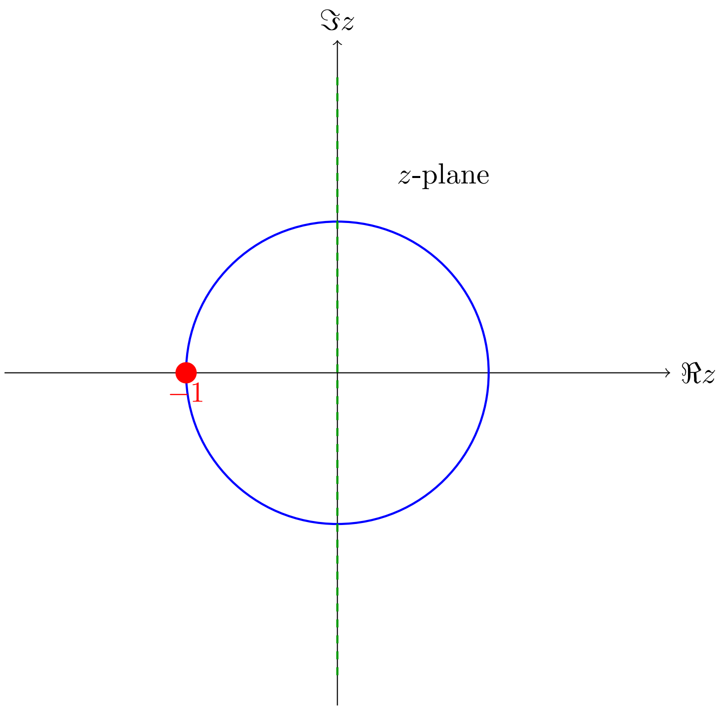
Description. This mapping is a shifted inversion. Let $w = z + 1$. Then

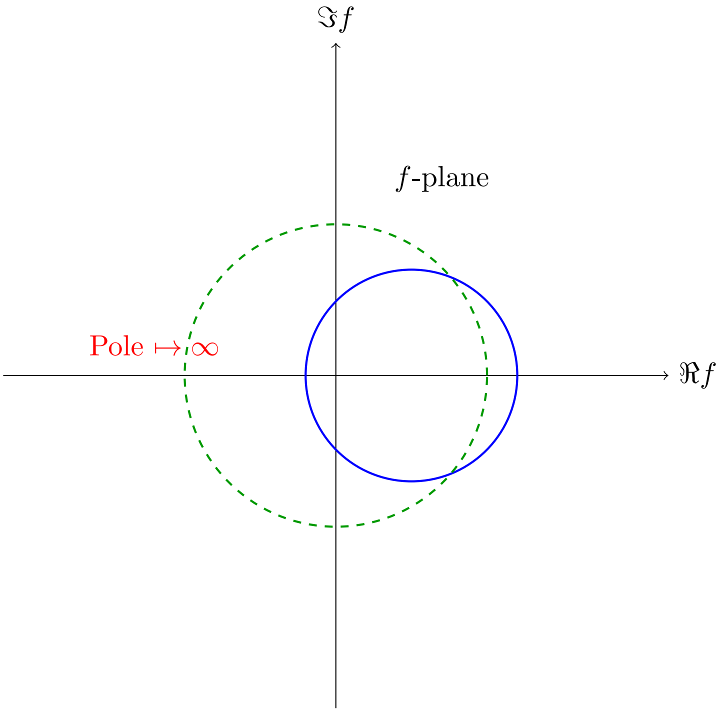
$$f(z) = \frac{1}{z+1} = \frac{1}{w}.$$

Thus the inversion is centered at $z = -1$ instead of $z = 0$. The function has a simple pole at $z = -1$. As $|z| \to \infty$, $f(z) \to 0$. Key geometric effects:

- Pole at $z = -1$ corresponds to $w = 0$ under translation.
- The unit circle $|z| = 1$ no longer maps to itself, but to another circle in the w-plane.
- Lines and circles not passing through $z = -1$ map to circles.
- Rays and bands relative to $z = -1$ are inverted accordingly.

Chart: Domain vs. Image under $1/(z + 1)$





Legend. Left: domain (z-plane) with unit circle (blue), pole at $z = -1$ (red), and imaginary axis (green dashed). Right: schematic image under $f(z) = 1/(z + 1)$: unit circle maps to another circle, imaginary axis maps to a circle through the origin, and $z = -1$ maps to infinity.

Example: $f(z) = \frac{1}{z^2}$

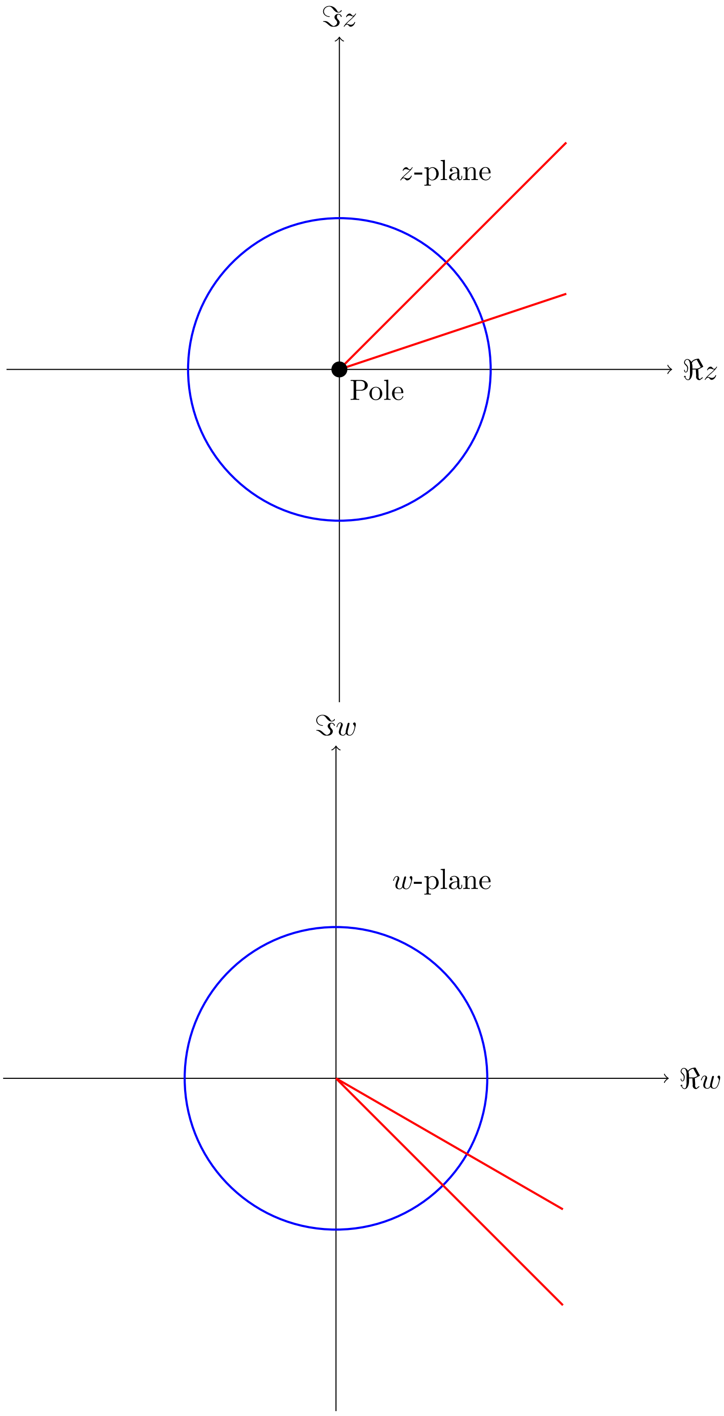
Description. For $z = re^{i\theta}$ we have

$$f(z) = \frac{1}{z^2} = \frac{1}{r^2} e^{-2i\theta}.$$

Thus the mapping $f(z) = 1/z^2$ has the following geometric effects:

- **Inversion of radii (squared):** $r \mapsto 1/r^2$.
- **Angle doubling and reflection:** $\theta \mapsto -2\theta$.
- The unit circle $|z| = 1$ is invariant (maps to itself).
- The interior of the unit disk $|z| < 1$ maps to the exterior $|w| > 1$, and vice versa.
- Rays at angle θ map to rays at angle -2θ.
- There is a pole of order 2 at $z = 0$.

Chart: Domain vs. Image under $1/z^2$



Legend. Blue: unit circle (invariant). Red: sample rays mapped under $\theta \mapsto -2\theta$. Black dot: pole of order 2 at $z = 0$.

Exercise 1.2 (Periodic Green's function for $-\psi'' + \psi = f$)

Suppose $f \in C^0(\mathbb{R})$ is 2π-periodic, i.e. $f(\theta) = f(\theta + 2\pi)$. For $\theta \in [0, 2\pi)$ define

$$\psi(\theta) := \int_0^\theta f(\alpha)\,\frac{\cosh(\pi - \theta + \alpha)}{2\sinh\pi}\,d\alpha + \int_\theta^{2\pi} f(\alpha)\,\frac{\cosh(-\pi - \theta + \alpha)}{2\sinh\pi}\,d\alpha, \tag{1}$$

and extend ψ to $\mathbb{R}$ by periodicity $\psi(\theta + 2\pi) = \psi(\theta)$.

Background / definition. It is convenient to write (**??**) using the periodic Green kernel

$$G(\theta,\alpha) := \frac{\cosh\bigl(\pi - |\theta-\alpha|\bigr)}{2\sinh\pi}, \qquad \theta,\alpha \in [0,2\pi). \tag{2}$$

Then (**??**) is equivalently

$$\psi(\theta) = \int_0^{2\pi} f(\alpha)\, G(\theta,\alpha)\, d\alpha, \tag{3}$$

i.e. $\psi = L^{-1} f$ for the periodic operator $L := -\frac{d^2}{d\theta^2} + 1$.

Solution. **(a) $\boldsymbol{\psi \in C^0}(\mathbb{R})$.** On $[0,2\pi)$ the map $(\theta,\alpha) \mapsto f(\alpha)\, G(\theta,\alpha)$ is continuous and bounded. By dominated convergence, $\theta \mapsto \psi(\theta)$ is continuous. The periodic extension is continuous since both f and G are 2π-periodic in each variable.

(b) First derivative and $\boldsymbol{\psi \in C^1}(\mathbb{R})$. Differentiate (**??**) by Leibniz' rule. Using $\partial_\theta \cosh(\pi - \theta + \alpha) = -\sinh(\pi - \theta + \alpha)$ and $\partial_\theta \cosh(-\pi - \theta + \alpha) = -\sinh(-\pi - \theta + \alpha)$, and noting that the boundary terms cancel because $\frac{\cosh\pi}{2\sinh\pi} - \frac{\cosh\pi}{2\sinh\pi} = 0$, we obtain

$$\psi'(\theta) = -\int_0^{\theta} f(\alpha)\, \frac{\sinh(\pi - \theta + \alpha)}{2\sinh\pi}\, d\alpha \;-\; \int_\theta^{2\pi} f(\alpha)\, \frac{\sinh(-\pi - \theta + \alpha)}{2\sinh\pi}\, d\alpha. \tag{4}$$

Again by dominated convergence, the right-hand side is continuous, hence $\psi \in C^1(\mathbb{R})$.

(c) Second derivative and the equation $\boldsymbol{-\psi'' + \psi = f}$. Differentiate (**??**) once more. Using $\partial_\theta \sinh(\pi - \theta + \alpha) = -\cosh(\pi - \theta + \alpha)$ and $\partial_\theta \sinh(-\pi - \theta + \alpha) = -\cosh(-\pi - \theta + \alpha)$, Leibniz' rule yields

$$\begin{aligned}
\psi''(\theta) &= -\frac{f(\theta)\sinh\pi}{2\sinh\pi} + \int_0^{\theta} f(\alpha)\, \frac{\cosh(\pi - \theta + \alpha)}{2\sinh\pi}\, d\alpha \\
&\quad - \frac{f(\theta)\sinh(-\pi)}{2\sinh\pi} + \int_\theta^{2\pi} f(\alpha)\, \frac{\cosh(-\pi - \theta + \alpha)}{2\sinh\pi}\, d\alpha \\
&= -f(\theta) + \psi(\theta).
\end{aligned}$$

Thus for all $\theta \in \mathbb{R}$, we have

$$-\psi''(\theta) + \psi(\theta) = f(\theta).$$

Since f is continuous, the display shows $\psi \in C^2(\mathbb{R})$.